WORKED SOLUTION

STEP II Specimen paper, Question 4

Pure • Hyperbolic Functions

STEP II Specimen paper, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s2-q4

The key idea

Every one of these identities is a statement about a single angle (or a single hyperbolic "angle"). Give the right-hand side a name, read off one function of it, and use the Pythagorean identities $1 + \tan^2\theta = \sec^2\theta$ and $1 + \sinh^2 u = \cosh^2 u$ to get the other. The only real care needed is with signs of square roots, which is exactly where the ranges of principal values come in — and, in the final part, where the identity stops being true.

Solution

Part (i)

Let $\theta = \tan^{-1}(\sinh x)$, so that $\tan\theta = \sinh x$ and, being a principal value, $-\frac{1}{2}\pi < \theta < \frac{1}{2}\pi$. Then

$$\sec^2\theta = 1 + \tan^2\theta = 1 + \sinh^2 x = \cosh^2 x.$$

On $(-\frac{1}{2}\pi, \frac{1}{2}\pi)$ we have $\cos\theta > 0$, so $\sec\theta > 0$; and $\cosh x > 0$ always. Taking the positive square roots, $\sec\theta = \cosh x$, and therefore

$$\sin\theta = \tan\theta\cos\theta = \frac{\sinh x}{\cosh x} = \tanh x.$$

Since θ lies in $[-\frac{1}{2}\pi, \frac{1}{2}\pi]$, the range of principal values of $\sin^{-1}$, this says precisely that $\theta = \sin^{-1}(\tanh x)$. Hence $\sin^{-1}(\tanh x) = \tan^{-1}(\sinh x)$ for every real x, as required.

? **Ask yourself** Why isn't $\sin\theta = \tanh x$ already the end of the argument?

Because $\sin\theta = \tanh x$ has infinitely many solutions θ. It is only because our θ lies in $[-\frac{1}{2}\pi, \frac{1}{2}\pi]$, where sin is one-to-one, that we may conclude θ *is* the principal value $\sin^{-1}(\tanh x)$. The same range also fixed the sign of $\sec\theta$.

Part (ii)

Let $u = \sinh^{-1}(\tan y)$, which is defined because $\tan y$ is finite for $-\frac{1}{2}\pi < y < \frac{1}{2}\pi$. Then $\sinh u = \tan y$, and part (i) with $x = u$ gives

$$\sin^{-1}(\tanh u) = \tan^{-1}(\sinh u) = \tan^{-1}(\tan y) = y,$$

where the last step uses $-\frac{1}{2}\pi < y < \frac{1}{2}\pi$, the range of principal values of $\tan^{-1}$. Taking sin of both sides, $\tanh u = \sin y$. As $|\sin y| < 1$ on this interval, $\tanh^{-1}(\sin y)$ is defined, and tanh is one-to-one, so $u = \tanh^{-1}(\sin y)$. That is,

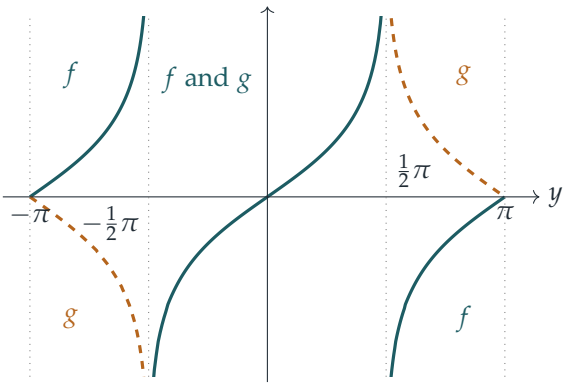
$$\sinh^{-1}(\tan y) = \tanh^{-1}(\sin y) \qquad \left(-\tfrac{1}{2}\pi < y < \tfrac{1}{2}\pi\right).$$

(Directly: $\cosh u = \sqrt{1+\tan^2 y} = \sec y$, positive because $\cos y > 0$, so $\tanh u = \tan y \cos y = \sin y$.)

The sketch Write $f(y) = \sinh^{-1}(\tan y)$ and $g(y) = \tanh^{-1}(\sin y)$. Both are undefined at $y = \pm\frac{1}{2}\pi$, where $\tan y$ is infinite and $\sin y = \pm 1$. Their behaviour elsewhere follows from the symmetries of tan and sin.

- Both are odd, and both vanish at $y = 0$ and $y = \pm\pi$.
- f has period π, because tan does. So on $(\frac{1}{2}\pi, \pi)$ it repeats its values on $(-\frac{1}{2}\pi, 0)$: it rises from $-\infty$ to 0.
- $\sin(\pi - y) = \sin y$, so g is symmetric about $y = \frac{1}{2}\pi$. On $(\frac{1}{2}\pi, \pi)$ it repeats its values on $(0, \frac{1}{2}\pi)$ in mirror image: it falls from $+\infty$ to 0.
- Differentiating, $f'(y) = \dfrac{\sec^2 y}{\sqrt{1+\tan^2 y}} = |\sec y|$ and $g'(y) = \dfrac{\cos y}{1-\sin^2 y} = \sec y$. So f is increasing on every interval where it is defined, while g increases on $(-\frac{1}{2}\pi, \frac{1}{2}\pi)$ and decreases outside it. Both have gradient 1 at the origin; at $y = \pi$, f has gradient 1 and g has gradient -1.

On $(-\frac{1}{2}\pi, \frac{1}{2}\pi)$ the two graphs are the same curve; on the outer intervals each is the reflection of the other in the horizontal axis.



Solid: $f(y) = \sinh^{-1}(\tan y)$. Dashed: $g(y) = \tanh^{-1}(\sin y)$ where it differs from f. The dotted lines $y = \pm\frac{1}{2}\pi$ are asymptotes.

The relationship for $\frac{1}{2}\pi < y < \pi$ The sketch suggests $f = -g$ there, and we can prove it. If $\frac{1}{2}\pi < y < \pi$ then $y - \pi$ lies in $(-\frac{1}{2}\pi, 0)$, so the identity just proved applies to $y - \pi$:

$$\begin{aligned}\sinh^{-1}(\tan y) &= \sinh^{-1}\big(\tan(y-\pi)\big)\\ &= \tanh^{-1}\big(\sin(y-\pi)\big)\\ &= \tanh^{-1}(-\sin y) = -\tanh^{-1}(\sin y),\end{aligned}$$

using $\tan(y-\pi) = \tan y$, $\sin(y-\pi) = -\sin y$ and the fact that $\tanh^{-1}$ is odd. So for $\frac{1}{2}\pi < y < \pi$,

$$\sinh^{-1}(\tan y) = -\tanh^{-1}(\sin y).$$

(By oddness the same relation holds on $-\pi < y < -\frac{1}{2}\pi$.)

Ask yourself Where does the argument of part (ii) break down for $\frac{1}{2}\pi < y < \pi$?

At the step $\tan^{-1}(\tan y) = y$: for y outside $(-\frac{1}{2}\pi, \frac{1}{2}\pi)$ the principal value is $y - \pi$, not y. Equivalently, in the direct argument $\sqrt{1+\tan^2 y} = |\sec y| = -\sec y$ there, because $\cos y < 0$, and that minus sign is exactly the one in the answer.

Checks

- At $y = 1$: $\sinh^{-1}(\tan 1) = \tanh^{-1}(\sin 1) \approx 1.2262$. At $y = 2$: $\sinh^{-1}(\tan 2) \approx -1.5235$ while $\tanh^{-1}(\sin 2) \approx 1.5235$, confirming the change of sign.
- Signs: for $\frac{1}{2}\pi < y < \pi$, $\tan y < 0$ but $\sin y > 0$, so the two expressions must have opposite signs; they cannot be equal.
- Part (i) at large x: both sides tend to $\frac{1}{2}\pi$, as they should.

We tested both identities at 10^5 random points, and the relation $f = -g$ on $\frac{1}{2}\pi < |y| < \pi$, numerically.

Common slips

- Writing $\sqrt{1+\tan^2 y} = \sec y$ without checking the sign of $\sec y$. This is the whole point of the final part.
- Concluding $\theta = \sin^{-1}(\tanh x)$ from $\sin\theta = \tanh x$ without saying why θ is the principal value.
- Drawing g with the same shape as f beyond $\frac{1}{2}\pi$: g inherits the symmetry of sin about $\frac{1}{2}\pi$, not the periodicity of tan.

Beyond STEP

The function $\mathrm{gd}(x) = \tan^{-1}(\sinh x) = \sin^{-1}(\tanh x)$ is the *Gudermannian* function. It links circular and hyperbolic functions without complex numbers, and part (ii) shows its inverse is $\sinh^{-1}(\tan y)$. It appears in the Mercator map projection, where it converts the vertical map coordinate into latitude. Inverse functions and their branches, which tripped up the identity here, become central when you study $\log z$ and $z^{1/2}$ for complex z. Needham's *Visual Complex Analysis* is a beautiful place to see this.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.